Université de Lille

Numerical Approximation of an Advection–Diffusion Problem

Finite Volume (FV) and Finite Element (FE) Methods

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Problem Statement & Objectives

The Problem :

$$\begin{cases} \partial_t u + \operatorname{div}(-\nabla u + \mathbf{V} u) = f & \text{in } (0, T] \times \Omega, \\ u = u_D & \text{on } (0, T] \times \Gamma_D, \\ (-\nabla u + \mathbf{V} u) \cdot \mathbf{n} = 0 & \text{on } (0, T] \times \Gamma_N, \\ u(0, \cdot) = u_0 & \text{in } \Omega. \end{cases}$$

Configuration:

- **Domain:** $\Omega = (0, 1)^2$.
- **Velocity:** $\mathbf{V} = (\eta, 0)$ and source $f = 0$.
- **Dirichlet (Γ_D):** $x = 0, 1$.
  $\Rightarrow u(0, y) = 1, \ u(1, y) = e^{\eta}$.
- **Neumann (Γ_N):** $y = 0, 1$ (Top/Bottom).

Objectives:

- Implement FV (Upwind, Scharfetter–Gummel) and FE ($\mathbb{Q}^1$) schemes.
- Study convergence, stability and long-time behavior.
- Analyze the influence of the Péclet number (η).

Methodology: Error Analysis & Metrics

Error norms

We compare the numerical solution u_h with the exact solution u_{ex}:

$$\|e\|_{L^2} = \left(\sum_{K \in \mathcal{T}_h} |K| \, |u_K - u_{ex}(x_K)|^2 \right)^{1/2}$$

$$\|e\|_{H^1} = \|\nabla(u_h - u_{ex})\|_{L^2(\Omega)}$$

Experimental Order of Convergence (EOC)

Computed on successive mesh refinements:

$$\text{Order} = \frac{\log(\text{Error}_k / \text{Error}_{k-1})}{\log(h_k / h_{k-1})}$$

Expected rates:

- Order 1: Upwind scheme
- Order 2: Scharfetter–Gummel, $\mathbb{Q}^1$ FEM (L^2)

FV: General Scheme & Implementation

Generic Scheme: Integration over control volume $K_{i,j}$:

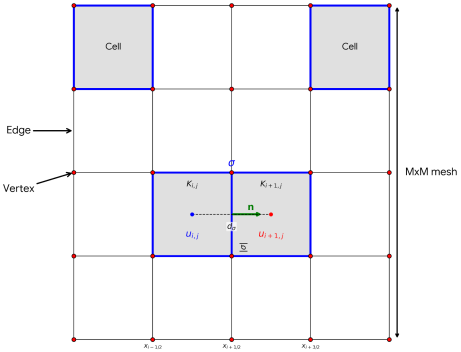
$$\frac{|K_{i,j}|}{\Delta t}(u_{i,j}^{n+1} - u_{i,j}^{n}) + \sum_{\sigma \in \mathcal{E}_{K_{i,j}}} \mathcal{F}_{K_{i,j},\sigma}^{n} = |K_{i,j}| f_{i,j}$$

Implementation Choices (Mesh Class):

- Object-oriented design: `Cell`, `Edge`, `Vertex`.
- Uniform Cartesian grid construction ($h = 1/M, M = 2^k$).
- Explicit handling of Boundary Conditions (Dirichlet/Neumann tags).

Numerical Flux Definition:

$$\mathcal{F}_{K,\sigma} = \mathcal{F}_{K,\sigma}^{L} \text{ (Diffusive)} + \mathcal{F}_{K,\sigma}^{T} \text{ (Convective)}$$



FV: Flux Strategies (Upwind vs SG)

1. Upwind Flux

Standard first-order approach:

$$\mathcal{F}^T_{K,\sigma} = |\sigma|\left(V^+ u_K + V^- u_L\right)$$

- Simple implementation.
- Condition: CFL

$$\Delta t \leq \frac{h^2}{4 + 2h\|\mathbf{V}\|_\infty}$$

2. Scharfetter-Gummel (SG)

Based on the Bernoulli function $B(z) = \frac{z}{e^z - 1}$:

$$\mathcal{F}_{K,\sigma} = \frac{|\sigma|}{d_\sigma}\left[B(-Vd_\sigma)u_K - B(Vd_\sigma)u_L\right]$$

- "Exact" for steady-state advection-diffusion.
- Handles high Péclet numbers robustly.

FV Implementation: Core Functions & Classes

1. **Mesh Data Structures** (`mesh_file.h`)
   - `Cell`: Stores geometric center $x_{i,j}$, area $|K|$, and neighbor pointers.
   - `Edge`: Stores normal $\mathbf{n}$, length $|\sigma|$, and distance d_σ.
   - `build_Cartesian(k)`: Generates the $2^k \times 2^k$ grid and connectivity.
2. **Time Loop** (`VF_scheme.cpp`)
   - Explicit time stepping.
   - Adaptive time step Δt based on CFL condition (Diffusive/Convective).
3. **Flux Solvers**

`update_solution` (Upwind)

Computes $\sum \mathcal{F}_{K,\sigma}$ using:

- **Diffusion:** Centered finite difference.
- **Advection:** Upwind based on sign of $V \cdot n$.

`update_solution_SG` (Scharfetter-Gummel)

Uses the helper `B_Bernoulli(z)`:

$$B(z) = z/(e^z - 1)$$

Updates solution using the exact flux formula derived from the 1D steady state.

FV Results: Upwind Convergence ($\eta = 1$)

Parameters: $\eta = 1$, $T = 0.5$. Error measured in L^2 norm.

k	h	Error L^2	Order	Time (s)
2	0.25	0.01270	0.97	0.00
3	0.13	0.00671	0.92	0.00
4	0.06	0.00353	0.93	0.03
5	0.03	0.00182	**0.95**	0.34
6	0.02	0.00093	0.97	5.52

Analysis:

- The method converges.
- The order stabilizes around **0.95**, confirming **first-order accuracy**.

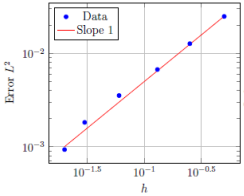


Figure 1: Log-log plot for Q7. The red line has slope 1.

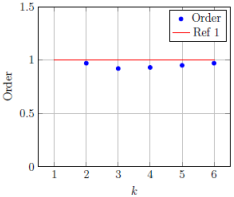


Figure 2: Order of convergence for Q7.

FV Results: Scharfetter-Gummel Convergence ($\eta = 1$)

Comparison: Same parameters ($\eta = 1,\ T = 0.5$).

k	h	Error L^2	Order	Time (s)
2	0.25	0.00079	2.60	0.00
3	0.13	0.00017	2.20	0.01
4	0.06	0.00004	2.08	0.12
5	0.03	0.00001	**2.04**	1.89
6	0.02	0.00000	**2.02**	29.94

Analysis:

- Significant improvement in accuracy.
- The scheme exhibits **second-order convergence** (≈ 2.02).
- Higher computational time due to stricter CFL ($\propto h^2$).

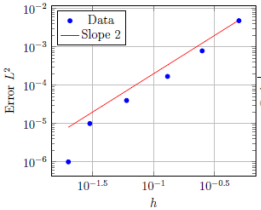


Figure 3: Log-log plot for SG scheme. Slope 2 for reference.

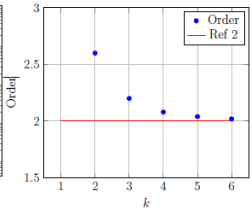


Figure 4: Order of convergence for SG scheme.

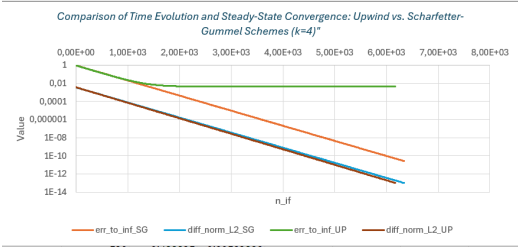
FV: Long-Time Behavior ($t \to \infty$)

Comparison of Steady-State Errors

We define the discrete steady state when residual $< 10^{-13}$.

Sch.	k	M	h	**Error** L^2	**Ord.**	n_1
SG	2	4	0.25	1.7×10^{-12}	-	493
	4	**16**	**0.06**	$\mathbf{2.6 \times 10^{-11}}$	**-1.9**	**6k**
	6	64	0.01	4.1×10^{-10}	-1.9	87k
Upw	2	4	0.25	1.4×10^{-2}	-	454
	4	**16**	**0.06**	$\mathbf{4.3 \times 10^{-3}}$	**0.9**	**6k**
	6	64	0.01	1.1×10^{-3}	1.0	87k

Table: Convergence results.



Observation

SG flux is **exact** for steady-state ($u^\infty = e^{\eta x}$). Error is machine precision.

FE: Weak Formulation & Properties

Variational Problem: Find $\tilde{u}(t) \in H^1_{0,D}(\Omega)$ such that:

$$\frac{d}{dt}\int_\Omega \tilde{u}(t)v\,dx + a(\tilde{u}(t), v) = \int_\Omega fv\,dx - a(\tilde{u}^D, v)$$

for all $v \in H^1_{0,D}(\Omega)$.

Bilinear Form $a(u, v)$: Derived using Green's formula (integration by parts on the divergence term):

$$a(u, v) = \int_\Omega \nabla u \cdot \nabla v\,dx - \int_\Omega (\mathbf{V}u) \cdot \nabla v\,dx$$

Properties:

- **Coercivity:** Since $\operatorname{div}\mathbf{V} = 0$ and $\mathbf{V}\cdot\mathbf{n} = 0$ on Γ_N, we have:

$$a(v, v) = \|\nabla v\|^2_{L^2}$$

- **Well-posedness:** The problem admits a unique solution (Lax-Milgram).

FE Implementation: Tensor Product Assembly

To optimize assembly on Cartesian grids, we use the tensor structure of $\mathbb{Q}^1$ basis functions: $\varphi_I(x,y) = \psi_i(x)\psi_j(y)$.

2D Elementary Matrices construction: We use Kronecker products ($\otimes$) of 1D reference matrices:

Stiffness Matrix (K_{el})

$$K_{el} = (K_{1D} \otimes M_{1D}) + (M_{1D} \otimes K_{1D})$$

$$K_{el} = \frac{1}{6}\begin{pmatrix} 4 & -1 & -1 & -2 \\ -1 & 4 & -2 & -1 \\ -1 & -2 & 4 & -1 \\ -2 & -1 & -1 & 4 \end{pmatrix}$$

Advection Matrix (C_{el})

$$C_{el} = \eta(C_{1D} \otimes M_{1D})$$

$$C_{el} = \frac{\eta h}{12}\begin{pmatrix} -2 & -1 & 2 & 1 \\ -1 & -2 & 1 & 2 \\ -2 & -1 & 2 & 1 \\ -1 & -2 & 1 & 2 \end{pmatrix}$$

Note: Matrices are independent of h for stiffness, and linear in h for advection.

FE Implementation: Assembly & Linear Solver

Library Used: `Eigen` (Sparse Algebra).

1. Elementary Matrices
(`compute_element_matrices`)

- **Optimization:** Matrices K_{el} and C_{el} are **hardcoded** (4x4) using the tensor product results (avoiding runtime quadrature loops).
- K_{el} is constant, C_{el} scales with ηh.

2. Global Assembly

- Loop over cells to fill `Eigen::Triplet`.
- Construction of `Eigen::SparseMatrix A`.

3. Boundary Conditions

- **Method:** Penalty Method.
- Large value (10^{14}) added to diagonal for Dirichlet nodes ($x = 0, x = 1$).

4. Solver & Error

- Direct Solver: `Eigen::SparseLU`.
- **Error Computation:** 2x2 Gauss Quadrature loop to integrate $\|u_h - u_{ex}\|^2$ accurately.

FE Results: Optimal Regime ($\eta = 1$ and 10)

Convergence study for moderate advection.

Table: Regime $\eta = 1$

k	h	$\|e\|_{L^2}$	Order	$\|e\|_{H^1}$	Order
3	0.125	2.15e-3	2.00	6.44e-2	0.99
4	0.062	5.39e-4	2.00	3.22e-2	0.99
5	0.031	1.35e-4	2.00	1.61e-2	0.99
6	**0.015**	**3.40e-5**	**2.00**	**8.06e-3**	**0.99**

Table: Regime $\eta = 10$

k	h	$\|e\|_{L^2}$	Order	$\|e\|_{H^1}$	Order
3	0.1250	452.15	1.94	16546.69	0.75
4	0.0625	113.37	1.99	8721.68	0.92
5	0.0312	28.33	2.00	4422.16	0.98
6	**0.0156**	**7.08**	**2.00**	**2218.93**	**0.99**

Outcome: Optimal convergence rates ($\mathcal{O}(h^2)$ for L^2, $\mathcal{O}(h)$ for H^1).
Observation: Rates are preserved (Order 2), but absolute errors increase drastically compared to $\eta = 1$ (7.08 vs 10^{-5}).

FE Results: Optimal Regime ($\eta = 1$ and 10)

Influence of increasing η on the error constant.

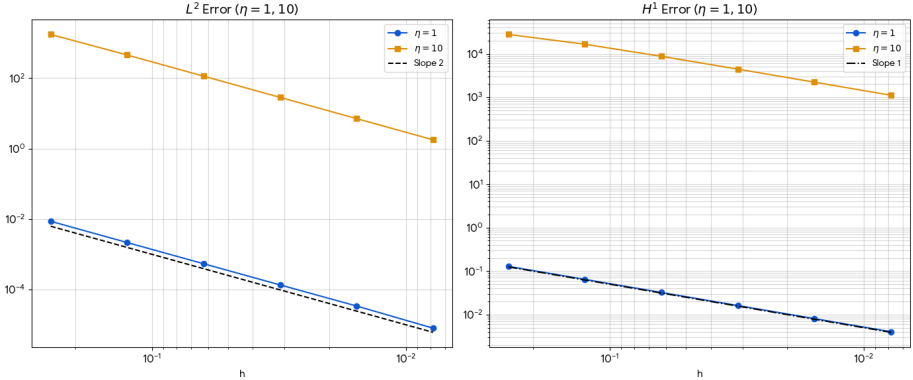
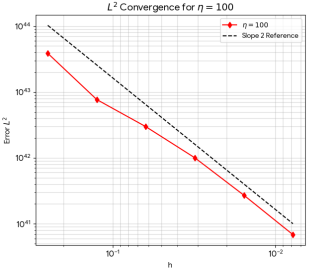


Figure: log-log scale of Error

FE Results: Pollution Effect ($\eta = 100$)

Context: Advection-dominated regime with Péclet number $Pe \gg 1$.

k	h	$\|e\|_{L^2}$	Order	$\|e\|_{H^1}$	Order
2	0.25	$3.86 \cdot 10^{43}$	-	$3.39 \cdot 10^{44}$	-
3	0.125	$7.67 \cdot 10^{38}$	2.33	$1.86 \cdot 10^{44}$	0.86
4	0.0625	$2.97 \cdot 10^{33}$	1.36	$1.49 \cdot 10^{44}$	0.32
5	0.0312	$1.00 \cdot 10^{28}$	1.57	$1.20 \cdot 10^{44}$	0.30
6	0.0156	$2.71 \cdot 10^{23}$	1.88	$7.69 \cdot 10^{43}$	0.65
7	0.0078	$6.83 \cdot 10^{18}$	1.98	$4.16 \cdot 10^{43}$	0.88



Pollution Effect Analysis

- Errors explode up to 10^{43} (exponential instability).
- **Cause:** Mesh not fine enough to resolve boundary layer ($h > 2/\eta$).
- Standard Galerkin unstable without stabilization (SUPG).

Bonus: Backward Euler Stability Analysis

Scheme: Implicit Backward Euler time discretization.
Theoretical Stability (Q24):

- The scheme is unconditionally stable (A-stable).
- No CFL condition required for stability (only for accuracy).

Discrete Exponential Convergence (Q25): We proved the error estimate:

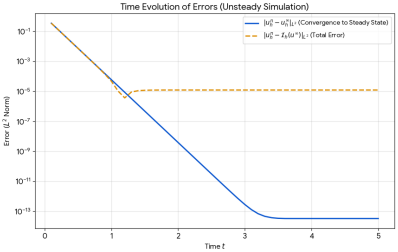
$$\|u_h^n - u_h^\infty\|_{L^2(\Omega)} \leq (1 + \frac{2\Delta t}{C_P^2})^{-n/2} \|u_h^0 - u_h^\infty\|_{L^2(\Omega)}$$

The discrete solution converges exponentially fast towards the discrete steady state.

Bonus: Numerical Time Convergence Results

Setup: $\eta = 1$, $k = 5$, $\Delta t = 0.01$, $T = 5.0$. We monitor $\|u_h^n - u_h^\infty\|_{L^2}$.

Iteration	Time (t)	$\|u_h^n - u_h^\infty\|_{L^2}$	$\|u_h^n - \mathcal{I}_h(u^\infty)\|_{L^2}$
10	0.1	3.49×10^{-1}	3.49×10^{-1}
50	0.5	7.36×10^{-3}	7.34×10^{-3}
100	1.0	5.92×10^{-5}	4.70×10^{-5}
150	1.5	4.76×10^{-7}	1.17×10^{-5}
200	2.0	3.83×10^{-9}	1.22×10^{-5}
300	3.0	2.79×10^{-13}	1.22×10^{-5}
500	5.0	3.22×10^{-14}	1.22×10^{-5}



Observation: Strict exponential decay until saturation at the spatial discretization error level ($\approx 10^{-5}$).

General Conclusion & R&D Perspectives

Finite Volumes

- **Upwind**
  - Unconditionally stable, monotone
  - Strong numerical diffusion $\Rightarrow$ Order 1 (≈ 0.95)
- **Scharfetter–Gummel (SG)**
  - Structure-preserving (Bernoulli exponential fit)
  - Order 2 (≈ 2.02), steady-state error $\sim 10^{-11}$
  - Best accuracy/robustness trade-off for $Pe \gg 1$

Finite Elements

- **Diffusion-dominated ($Pe_h < 1$)**
  - Standard Galerkin $\mathbb{Q}^1$ optimal
  - $\mathcal{O}(h^2)_{L^2}$, $\mathcal{O}(h)_{H^1}$
- **Advection-dominated ($Pe_h \gg 1$)**
  - Maximum principle violated $\Rightarrow$ oscillations
  - Severe pollution if layers unresolved ($h > 2/\eta$)

R&D Recommendation

For $Pe \gg 1$, standard Galerkin requires stabilization (SUPG/VMS). The **Scharfetter–Gummel** FV scheme provides a robust, parameter-free industrial alternative.

Thank you for your attention!

Questions?